

POCKET PRODUCER: SESSION-CONDITIONED RETRIEVAL AS USER-CONTROLLED PROPOSAL

Chuling Li

University of Nottingham

scy114@nottingham.ac.uk

ABSTRACT

Music continuation need not coincide with audio similarity: the loop that best matches a session is not necessarily the one that best moves it forward. Pocket Producer is a working prototype integrated into the Audiotool DAW and designed around this distinction. It fingerprints the live session, ranks candidate loops using pack co-membership as cold-start weak supervision, and presents the results through an inspectable and reversible proposal workflow. Offline studies show that the apparent advantage of the learned ranker is sensitive to label construction, negative regime, and embedding representation. These observations motivate a design stance in which retrieval scores generate *proposals* that musicians can inspect, audition, insert, and undo, rather than final judgments about what should come next.

1. INTRODUCTION

When a musician returns to an unfinished loop-based project, the question is not “what sounds similar?” but “what would be useful as the next step?” Loop compatibility and sample retrieval [1, 2], stem coherence [3, 4], session-conditioned generation [5, 6], and human-in-the-loop co-creation [7] all address parts of this problem. We define **continuation utility** as the property that a candidate helps the musician make progress in the current session; unlike pairwise audio similarity, it is directional and conditioned on the evolving session context.

We contribute: (1) a problem formulation that separates continuation utility from similarity and compatibility; (2) a working prototype in a production DAW, engineered for revisability: live session fingerprinting, per-signal evidence, single-transaction insertion whose undo also retracts the candidate’s tags from the ranking context, and feedback logging; (3) a tiered negative-regime evaluation that characterizes the sensitivity of weakly supervised ranking to label construction, negative regime, and embedding representation.

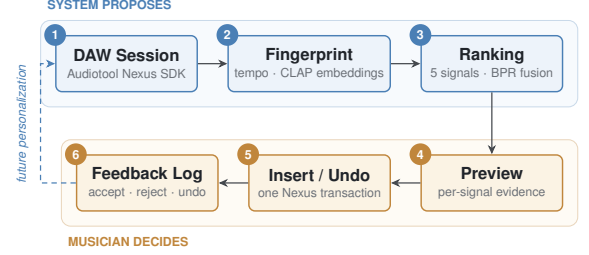


Figure 1. Pocket Producer pipeline: the system proposes (steps 1–3); the musician decides (steps 4–6).

2. SYSTEM

2.1 Session Fingerprinting

The live Audiotool session (Figure 1) is read via the Nexus SDK, which exposes the project’s tracks, regions, and referenced audio samples. The system extracts the project tempo from the document metadata and collects up to eight unique audio samples referenced by regions in the arrangement. Each sample is embedded with CLAP [8]; the resulting vectors form the session’s timbral fingerprint. When no session audio is available (empty project or no sample-backed regions), the system falls back to non-audio signals only.

2.2 Multi-Signal Ranking

Five signals score each FSLD [9] candidate (the offline evaluation corpus; Section 2.4 describes the deployed candidate library) against the session fingerprint: (i) mean-pooled and (ii) max-pooled CLAP cosine similarity to session samples, (iii) tempo proximity, (iv) key compatibility, and (v) tag Jaccard overlap between candidate tags and tags from the session’s audio fragments. Writing $\phi(s, x) \in \mathbb{R}^5$ for these signals given session s and candidate x , the fusion scores each candidate as $\sigma(w^\top \phi(s, x))$ (σ is monotone, so ranking follows $w^\top \phi$), with w trained by BPR [10] on pack co-membership weak labels (Section 3):

$$\mathcal{L}_r(w) = -\mathbb{E}_{\mathcal{D}_r} [\log \sigma(w^\top (\phi(s, x^+) - \phi(s, x^-)))], \quad (1)$$

where the expectation is over triples (s, x^+, x^-) and r indexes the negative regime from which x^- is drawn. A hand-tuned *rules* baseline over non-audio session signals serves as the default ranker; the fusion is selectable and falls back to rules when no session audio is available.



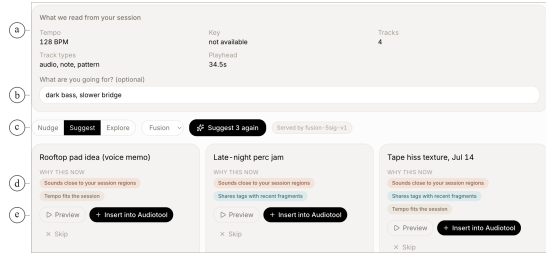


Figure 2. Continuation panel (interface rendering with illustrative data). (a) session fingerprint summary; (b) text intent; (c) intensity and ranker; (d) evidence chips; (e) audition, insert, skip. Candidates come from the musician’s captured fragments.

2.3 Proposal Workflow

The proposal workflow keeps creative authority with the musician. Each candidate carries per-signal evidence chips (e.g., timbral closeness, tempo fit, shared tags) so the musician can judge whether those reasons apply. Candidates can be previewed, inserted via a single Nexus transaction, and undone, restoring the arrangement; undo also retracts the candidate’s tags from the ranking context. Events are logged for future personalization (not yet evaluated).

2.4 Deployment Configuration

For deployment, we adapt the same five-signal ranker to the musician’s fragment library with four differences: MS-CLAP embedding space, chosen for licensing and CPU inference (offline evaluation uses LAION-CLAP Music); source-weighted BPR (hard_similar $\times 2.0$, mean over 5 seeds); key constant zero online (Audiotool documents lack key-signature metadata); tag context drawn from session fragments rather than FSLD pack siblings.

3. COLD-START WEAK SUPERVISION

At launch no interaction data exists, so the fusion is trained on proxy supervision that predates any user: pack co-membership. From the FSLD [9] we kept only sounds with a true Freesound pack (1,797 items, 605 packs), discarding items grouped only by uploader; each pack is one *source*, the grouping unit for all splits. Each eligible pack with at least three items yields one leave-one-out query per item, with same-pack siblings as positives and cross-pack items as negatives in three difficulty tiers (*easy*: tempo/key mismatch; *hard_tempo_key*: compatible tempo and key; *hard_similar*: highest cosine to the positive from another pack). We use a fixed 121-source held-out split (seed 20260810). Development uses only the 484 train sources; the 171 with ≥ 3 items form queries (981 examples), the rest contribute negatives only. LAION-CLAP Music [11] (512-dim) provides the embedding backbone.

We report pairwise ranking accuracy: the fraction of positive-negative pairs in which the positive scores higher (ties count 0.5). On the train side (mean over 5 source-grouped splits, each training on $\sim 75\%$ of train sources for 300 epochs, no early stopping), frozen cosine reaches

Negative regime	LAION-CLAP	MS-CLAP, sw-BPR
easy	+0.2 [−0.5, +0.9]	+0.1 [−1.2, +1.4]
hard_tempo_key	+0.0 [−0.8, +0.5]	−0.1 [−2.2, +2.0]
hard_similar	+2.1 [−1.0, +5.3]	+9.9 [+4.4, +15.4]
overall	+0.6 [−0.4, +1.6]	+2.6 [+1.0, +4.2]

Table 1. Held-out fusion—cosine delta (pp, mean over 5 seeds) with bootstrap 95% CIs (10,000 source-level resamples), per negative regime and embedding/training configuration (sw-BPR: source-weighted BPR). The apparent gain concentrates in hard_similar; the 7.8 pp cross-configuration swing shows sensitivity to embedding and training choices.

0.929 overall and 0.8136 on hard_similar; five-signal fusion reaches 0.925 and 0.8133. Under pack-only labels the fusion advantage is gone: -0.03 pp on hard_similar, and fusion trails cosine on hard_tempo_key by 2.1 pp on average (negative in all 5 splits).

4. OBSERVATIONS

Three lines of evidence characterize the sensitivity of the cold-start proxy. **Label sensitivity.** An earlier corpus treated same-uploader items as co-members (26% of examples, 949 sources); fusion appeared $+4.0$ pp over frozen cosine on hard_similar. Restricting to true pack co-membership erases the gain (Section 3). **Regime dependence.** Leave-one-signal-out ablation on the dev set shows tempo helps on hard_similar (-0.96 pp when removed) but hurts on hard_tempo_key ($+0.68$ pp); signals switch sign across negative regimes. In terms of Eq. (1), the learned optimum w^* depends on the training mixture over regimes $\mathcal{D}_r$ rather than defining a regime-invariant ranking. **Held-out uncertainty.** After correcting an initial training-protocol mismatch, we retrained on all 484 train sources and re-evaluated on the held-out split; we therefore treat Table 1 as exploratory. Under LAION-CLAP the hard_similar advantage is $+2.1$ pp with a bootstrap 95% CI crossing zero; under MS-CLAP with source-weighted BPR (a sensitivity re-analysis on the same split) it is $+9.9$ pp, a 7.8 pp cross-configuration swing. The 7.8 pp swing further illustrates configuration sensitivity.

5. CONCLUSION

Pocket Producer operationalizes session-conditioned retrieval as user-controlled proposal in the Audiotool DAW. Because conclusions drawn from the cold-start proxy shift with label, regime, and representation choices, ranking scores are surfaced as reversible proposals rather than final judgments. Future work includes evaluating creative outcomes, personalization from logged feedback, and regime-robust ranking objectives. A demonstration video accompanies this submission.

6. ACKNOWLEDGEMENTS

The author thanks the Audiotool team for the Nexus SDK and the FSLD creators [9].

7. AI USAGE STATEMENT

Claude Code (Anthropic) assisted with engineering, scripting, literature search, result auditing, and manuscript editing. OpenAI Codex assisted with claim auditing and prose review. All experimental design, analysis decisions, claims, and final content are the author's responsibility.

8. REFERENCES

- [1] B.-Y. Chen, J. B. L. Smith, and Y.-H. Yang, "Neural loop combiner: Neural network models for assessing the compatibility of loops," in *Proceedings of the 21st International Society for Music Information Retrieval Conference (ISMIR)*, 2020.
- [2] S. Lattner, "SampleMatch: Drum sample retrieval by musical context," in *Proceedings of the 23rd International Society for Music Information Retrieval Conference (ISMIR)*, 2022.
- [3] R. Ciranni, G. Mariani, M. Mancusi, E. Postolache, G. Fabbro, E. Rodolà, and L. Cosmo, "COCOLA: Coherence-oriented contrastive learning of musical audio representations," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- [4] A. Riou, A. Gagnere, G. Hadjeres, S. Lattner, and G. Peeters, "Zero-shot musical stem retrieval with joint-embedding predictive architectures," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- [5] Q. He, G. Xia, and Z. Wang, "TOMI: Transforming and organizing music ideas for multi-track compositions with full-song structure," in *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR)*, 2025.
- [6] J. Nistal, M. Pasini, C. Aouameur, M. Grachten, and S. Lattner, "Diff-A-Riff: Musical accompaniment co-creation via latent diffusion models," in *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR)*, 2024.
- [7] A. C. Elkins, S. Singh, A. Kieback, S. Blankenship, U. P. Amadasun, and A. Chadha, "DAWZY: A new addition to AI powered 'human in the loop' music co-creation," in *NeurIPS Workshop on AI for Music*, 2025.
- [8] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, "CLAP: Learning audio concepts from natural language supervision," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [9] A. Ramires, F. Font, D. Bogdanov, J. B. L. Smith, Y.-H. Yang, J. Ching, B.-Y. Chen, Y.-K. Wu, W.-H. Hsu, and X. Serra, "The Freesound Loop Dataset and annotation tool," in *Proceedings of the 21st International Society for Music Information Retrieval Conference (ISMIR)*, 2020.
- [10] S. Rendle, C. Freudenthaler, Z. Gantner, and L. Schmidt-Thieme, "BPR: Bayesian personalized ranking from implicit feedback," in *Proceedings of the 25th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2009.
- [11] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, "Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation," in *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.